

Microtubule Quantum Effects and Thalamocortical Integration: A Multiscale Hypothesis for Consciousness

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Abstract

The hard problem of consciousness remains unresolved within current neuroscience. Here I propose a **multiscale hypothesis** in which consciousness depends on an interaction between subcellular biophysics and large-scale thalamocortical dynamics. Neuronal microtubules are treated as **candidate quantum-sensitive substrates**: tubulin architectures exhibit ultraviolet superradiance [1]; microtubules support long-range electronic energy migration *in vitro* [2]; general anaesthetics alter these energy-transfer properties [3]; and isolated brain microtubules display intrinsic electrical oscillations with a 39 Hz peak under stabilised conditions [4–7]. Three quantitative plausibility simulations support the biophysical feasibility of this framework. At the systems level, thalamocortical circuitry—especially the centromedian–parafascicular complex—remains the best-supported macroscale substrate for conscious state regulation [8–10]. The central claim is limited and testable: **microtubule-level quantum or electrodynamic processes may modulate neuronal dynamics that are subsequently integrated by thalamocortical circuits into conscious state**, but the crucial mechanistic bridges remain unproven. **Critically, the causal link between Tier 1 (quantum-optical) and Tier 2 (electrodynamic oscillation) phenomena has not been experimentally demonstrated; the present framework treats them as co-localised observations requiring a bridge experiment (Section 7, Experiment 6).** Seven falsifiable experiments are proposed.

Data & Code Availability: All simulation code, parameters, random seeds, and supplementary figures are publicly available at github.com/ozyurte/microtubule-thalamocortical-consciousness.
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1 Introduction

1.1 The Explanatory Gap

Consciousness science faces a structural impasse. Classical neuroscience has produced detailed maps of neural correlates of consciousness—specific brain regions, circuits, and oscillatory patterns that covary with conscious states—yet it has not explained why these physical processes are accompanied by subjective experience. The explanatory gap between objective neural description and first-person phenomenology, formalised by Chalmers (1995) as the “hard problem,” persists because the standard computational framework treats neurons as classical information processors and consciousness as an emergent property of sufficient computational complexity [11].

1.2 Three Anomalies Motivating a Sub-cellular Investigation

This explanatory gap is not merely philosophical. It manifests in concrete clinical and pharmacological puzzles that classical models struggle to resolve.

The anaesthesia mechanism problem. General volatile anaesthetics—chemically diverse molecules ranging from single-atom noble gases (xenon) to complex halogenated ethers (sevoflurane)—all produce reversible unconsciousness, yet no single membrane receptor target fully explains their shared phenomenology across structurally diverse agents. The Meyer–Overton correlation, established over a century ago, demonstrates that anaesthetic potency scales linearly with solubility in non-polar lipid-like media rather than with affinity for any specific receptor protein [3]. Classical receptor-based pharmacology cannot explain why structurally identical non-anaesthetic analogues

with equal or superior lipid solubility completely fail to produce unconsciousness and in some cases induce convulsions.

Covert consciousness. Advances in clinical neurology have revealed that approximately 15–25% of patients diagnosed with unresponsive wakefulness syndrome (UWS) retain covert conscious awareness detectable only through functional neuroimaging or electrophysiological complexity measures [12, 13]. These patients with cognitive motor dissociation (CMD) possess intact thalamocortical integration circuits despite complete motor output failure, demonstrating that consciousness depends on specific circuit architectures rather than on general brain computational capacity.

Isotope-specific neural effects. Lithium-6 and lithium-7, stable isotopes that are chemically and electrically identical but differ in nuclear spin ($I = 1$ and $I = 3/2$ respectively), produce measurably different neurological effects in animal models [14–16]. Classical mass-action pharmacology, which depends solely on electronic structure and binding affinity, cannot account for isotope-dependent neural modulation driven by nuclear spin properties.

The connectome completeness problem. The completion of the adult *Drosophila* whole-brain connectome (FlyWire; Dorkenwald et al. 2024) provides the first opportunity to test exhaustively whether synaptic wiring diagrams alone suffice to explain brain function [52]. Simulations using biophysical leaky-integrate-and-fire models on this map reproduce substantial sensorimotor logic, yet identify systematic gaps: anatomical connectivity does not equate to functional efficacy, and many strong inhibitory connections remain “functionally silent” during behaviour. These findings do not prove that intracellular dynamics are necessary, but they delineate the empirical boundary of what synaptic-only models can currently explain—and identify the space in which subcellular modulatory mechanisms, including microtubule electrodynamics, could contribute.

1.3 Scope and Framing

This paper does **not** claim that consciousness has been localised to microtubules, nor that quantum coherence in microtubules has been shown to generate subjective experience. Instead, it argues for a narrower proposition: that a classically incomplete account of consciousness may justify investigation of microtubules as a biologically plausible modulatory layer operating below synaptic network dynamics, while thalamocortical circuits remain the best-supported macroscale substrate for conscious state regulation.

Evidence supporting this proposition is currently uneven: the thalamocortical literature is clinically and experimentally strong, whereas the microtubule literature is intriguing but still largely indirect with respect to conscious experience itself.

The framework is organised into three tiers of descending evidential strength. Tier 3 (thalamocortical

integration) is the strongest. Tier 1 (microtubule quantum properties) and Tier 2 (microtubule electrodynamics) present candidate mechanisms that require further *in vivo* validation. Three computational plausibility simulations (Section 6) provide quantitative bounds on signal magnitude, intrinsic frequency, and network-level entrainment. The philosophical extension (universal consciousness field) is explicitly separated as untestable speculation. This framework should also be distinguished from the Orchestrated Objective Reduction (Orch OR) hypothesis of Hameroff and Penrose [17], which shares the structural focus on microtubules but differs in three fundamental respects. First, Orch OR posits that consciousness arises from quantum gravitational state reduction (objective reduction) occurring within microtubule tubulin superpositions at specific mass-energy thresholds; the present hypothesis makes no claim about quantum gravity, wave-function collapse mechanisms, or any specific quantum-to-classical transition and is fully compatible with standard decoherence if quantum coherence proves functionally irrelevant *in vivo*. Second, Orch OR treats microtubules as the primary *locus* of conscious experience—the site where proto-conscious moments are “orchestrated”; the present framework treats microtubules as a *modulatory layer* subordinate to thalamocortical integration, which remains the macroscale substrate for conscious state regulation (Tier 3). Third, Orch OR predicts temporally discrete “conscious events” at intervals determined by $E = \hbar/t$ (the Diósi–Penrose criterion); the present hypothesis predicts only that microtubule electrodynamic properties may continuously modulate neuronal signalling parameters—a claim testable by classical electrophysiology (Experiments 1, 6, and 7) without requiring any quantum gravity framework. In summary, the present proposal is more conservative than Orch OR: it asks whether microtubules modulate neural dynamics, not whether they generate consciousness through quantum gravitational collapse. It should be emphasised that the Tier 1 $\rightarrow$ Tier 2 transduction pathway (UV/exciton $\rightarrow$ 39 Hz ionic oscillation) remains entirely hypothetical; no experiment has yet linked these two independently supported phenomena causally.¹

2 Tier 1: Candidate Quantum-Sensitive Substrate—Microtubule Biophysics

Neuronal microtubules possess structural and optical properties that make them plausible candidates for nontrivial subcellular information dynamics. Experimental work supports collective ultraviolet optical effects in tubulin assemblies and long-range electronic

¹For the most recent defence and extension of Orch OR, including a quantum solution to the phenomenal binding problem and the epiphenomenalism problem, see Wiest (2025) [46]. The present framework does not engage with these philosophical extensions, as they depend on the objective reduction mechanism that this paper explicitly sets aside.

energy migration in microtubules *in vitro*. However, these findings do **not** yet establish cognition-relevant quantum information processing in living brain tissue.

2.1 Structural and Electromagnetic Properties of Tubulin

Microtubules are hollow cylindrical polymers composed of α - β tubulin heterodimers arranged in a 13-protofilament helical lattice with an outer diameter of approximately 25 nm and an inner lumen diameter of approximately 15 nm. Each tubulin heterodimer carries an intrinsic electric dipole moment of approximately 1750 Debye—nearly three orders of magnitude greater than water (1.85 Debye)—and a net negative charge of approximately $-23e$ [3]. When assembled into the crystalline microtubule lattice, these dipoles create a macroscopic polarisable structure with sensitivity to electromagnetic fields, static and alternating electric fields, and local ionic environment fluctuations.

2.2 Tryptophan Superradiance Networks

Each tubulin dimer contains eight tryptophan residues whose indole ring transition dipoles participate in ultraviolet-range excitonic interactions. When tubulin assembles into microtubules and microtubules bundle into hexagonal packing arrangements characteristic of neuronal dendrites, these tryptophan chromophores form extended “mega-networks” comprising over 10^5 UV-excited transition dipoles.

Babcock and Kurian (2024) demonstrated that these tryptophan mega-networks exhibit collective quantum optical behaviour consistent with superradiance—a cooperative emission phenomenon in which the system radiates as a single quantum mechanical entity rather than as independent molecules [1]. Fluorescence quantum yield (QY) increased with hierarchical structural organisation (isolated tryptophan $\rightarrow$ tubulin dimer $\rightarrow$ microtubule assembly), with the enhancement scaling according to superradiant eigenmode predictions in the single-excitation manifold.

Critically, these collective effects persisted under thermal equilibrium conditions at physiological temperatures, with the superradiant dipole strength distributing across nearby excitonic states to confer robustness against the $\sim 200 \text{ cm}^{-1}$ static disorder characteristic of the cellular environment. The system simultaneously supports “bright” superradiant states with radiative lifetimes of hundreds of femtoseconds and “dark” subradiant states capable of storing excitonic energy on longer timescales [1]. Celardo et al. (2019) independently confirmed the existence of superradiant excitonic states in microtubule tryptophan networks using open quantum systems methods [18]. Gassab and Craddock (2026) provide the most current synthesis, confirming that both bright and dark exciton channels coexist in the cytoskeletal lattice [19].

Limitations: These results demonstrate collective quantum optical behaviour in tubulin architectures under laboratory conditions. They do not yet establish that neurons exploit UV excitations for signalling at timescales relevant for cognition, nor do they demonstrate information-bearing quantum coherence in living brain tissue. The bridge from fluorescence quantum yield measurements to functional neural information processing remains to be constructed.

2.3 High-Q Cavity Shielding and Decoherence Protection

A persistent objection to quantum biological models is the decoherence problem: thermal noise should collapse quantum superpositions within femtoseconds ($\sim 10^{-13} \text{ s}$). However, the microtubule lumen contains ordered interfacial water layers (“structured water”) whose properties differ from bulk cytoplasmic water. The interaction between high-dipole tubulin subunits and these ordered water layers creates conditions modelled as a high quality factor (high- Q) quantum electrodynamics (QED) cavity [17].

Several complementary mechanisms may extend coherence times: topological protection from crystalline lattice geometry, electromagnetic shielding within deep hydrophobic protein pockets, ordered interfacial water layers reducing thermal coupling, and metabolically-driven Fröhlich condensation. Computational modelling has revised estimated coherence times upward from femtoseconds to the microsecond–millisecond regime (10^{-6} to 10^{-3} s), overlapping with rapid neurophysiological process timescales.

Limitations: These coherence time estimates are derived from computational models and indirect fluorescence measurements, not from direct *in vivo* quantum state tomography. The existence of biologically functional quantum coherence in intact brain tissue remains an open question.

The most influential quantitative objection remains Tegmark’s estimate of a decoherence time of $\sim 10^{-13} \text{ s}$ for microtubule superpositions [39]. Hagan, Hameroff, and Tuszynski [40] responded that Tegmark’s calculation assumed an unrealistically large superposition separation distance and thermal equilibrium conditions inappropriate for metabolically active biological systems. The present hypothesis does not depend on resolving this debate: Tiers 1 and 2 are compatible with entirely classical biophysical mechanisms if quantum coherence proves irrelevant *in vivo* (Section 1.3). However, the experimental observations of room-temperature superradiance [1] and electronic energy migration [2] demonstrate that *some* collective quantum-optical effects survive thermal noise in microtubule preparations, even if their functional relevance to neural signalling remains unestablished.

2.4 Anaesthetic Interactions with Microtubule Quantum Properties

Anaesthesia remains one of the most relevant empirical entry points for this framework, but the claim must be stated carefully.

Volatile anaesthetic molecules partition into hydrophobic pockets within tubulin’s interior through weak van der Waals London dispersion forces. These pockets house dense, highly polarisable π -electron resonance clouds supporting excitonic mobility and resonant energy transfer throughout the microtubule lattice [3].

Kalra et al. (2023) demonstrated that anaesthetics reduce electronic energy migration in microtubules *in vitro*, providing direct evidence that general anaesthetics can alter microtubule energy-transfer properties [2]. Craddock et al. (2017) showed through quantum chemical modelling that clinically used anaesthetic gases universally down-shifted a collective dipole oscillation mode at 613 THz in tubulin, with the magnitude of frequency shift directly proportional to clinical potency (MAC value). Non-anaesthetic structural analogues bound to the same sites but did not shift the 613 THz mode [3].

2.4.1 In Vivo Behavioural Validation: Epothilone B Studies

Khan, Wiest et al. (2024) tested whether microtubule stabilisation alters anaesthetic sensitivity in 12 male Long-Evans rats [20]. Rats pre-treated with 0.75 mg/kg epothilone B (a brain-penetrant microtubule stabiliser) took an average of 69 seconds longer to reach loss of righting reflex under 4% isoflurane ($p = 0.0016$; Cohen’s $d = 1.9$). A 2026 mouse replication study confirmed the direction of the effect (Cohen’s $d = 0.8$) [21], and a complementary *Tetrahymena* study showed epothilone B conferred resistance to isoflurane-induced swimming slowdown in a single-celled organism [22].

These results justify treating microtubules as a potentially relevant intracellular target of at least some anaesthetic effects. They do **not** yet establish that anaesthetic-induced unconsciousness is primarily caused by “quenching” a microtubule quantum state, nor do they displace established receptor-level and systems-level mechanisms. A parsimonious alternative interpretation is binding-site competition, in which epothilone B partially occupies or sterically alters the isoflurane-binding region on tubulin, rather than global quantum state disruption.

An important caveat is that the direction of the epothilone effect may depend on dosing regimen. Li et al. (2025) found that chronic daily administration of microtubule-modulating drugs over two weeks altered isoflurane sensitivity in mice in a drug-specific manner: epothilone D and vinblastine *increased* sensitivity (EC_{50} decreased from ~ 0.97 to ~ 0.75 and ~ 0.74 , respectively), while paclitaxel caused a modest *decrease* in sensitivity ($EC_{50} \sim 1.05$ vs. ~ 0.98 for saline) [23].

These chronic results suggest that sustained microtubule modulation triggers compensatory adaptations, complicating simple mechanistic interpretation. The small sample sizes ($n = 12$ rats in the original study, plus mouse and *Tetrahymena* cohorts) and the availability of a classical pharmacological explanation mean that these results should be considered preliminary rather than decisive.

3 Tier 2: Candidate Electrodynamic Transduction Layer—Microtubule Electrical Oscillations

Isolated brain microtubules and microtubule bundles have been reported to generate self-sustained electrical oscillations *in vitro*, including a prominent 39 Hz component under some stabilised conditions. These findings justify the hypothesis that microtubules may contribute to intracellular electrodynamics, but they do **not** yet show that microtubules drive cortical gamma rhythms *in vivo*.

3.1 Direct Electrophysiological Measurements

Using modified loose patch-clamp techniques under symmetrical ionic conditions at 0 mV holding potential (eliminating external voltage driving forces), Cantero et al. demonstrated that isolated rat brain microtubules generate continuous, self-sustained alternating currents [4, 5]. Single microtubules radiated electrical power on the order of 10^{-18} W with an intrinsic electromotive force of $V_{\text{eff}} = 0.63 \mu\text{V}$ under zero-gradient conditions.²

Non-stabilised microtubules exhibited complex broadband oscillatory responses with multiple overlapping frequency peaks at approximately 7, 13, 28, 39, 47, 90, 140, and 147 Hz, and two distinct conductance regimes: a “high” state ($680 \pm 53 \text{ nS}$) and a “low” state ($13.4 \pm 2.3 \text{ nS}$), representing 258% conductance variation [4].

Paclitaxel-stabilised microtubules showed dramatically different behaviour: the broad spectrum collapsed into a single dominant fundamental peak at exactly 39 Hz with a minor harmonic near 74 Hz. These 39 Hz limit-cycle oscillations were continuous, self-sustained, and independent of applied voltage [5].

²The Discussion section of Cantero and Cantiello (2023) states “63 μV ,” but this is inconsistent with the explicit calculation in the Results section of the same paper, which yields $0.063 \times 10^{-5} \text{ V} = 0.63 \mu\text{V}$, and with the subsequent Nernst equation application ($\Delta C = 3.38 \mu\text{M}$), which is consistent only with the lower value. No published erratum exists as of March 2026. The present paper uses the mathematically verified value of 0.63 μV .

3.2 Physical Mechanism: Nanopore Gating and Ionic Solitons

The tubulin lattice contains regular nanopores at interdimer and intradimer interfaces—conical channels approximately 4 nm long with apertures of 0.7–1.0 nm—permeable to cations [7, 24]. Condensed counterion layers on inner and outer microtubule surfaces create two coupled, asymmetric, nonlinear electrical transmission lines. Counterion condensation enables ionic solitons—localised, non-dissipating charge waves—that propagate longitudinally along the lattice. Synchronised nanopore gating drives cyclical ionic current shifts generating the macroscopic alternating current at 39 Hz.

Marucho et al. (2025) independently developed multi-scale electrokinetic models successfully replicating the coupled nonlinear transmission line behaviour [7]. Their analytical prediction yields a soliton frequency of $f = 45.77$ Hz; the experimental value from Cantero et al. is 39.1 Hz. The soliton damping parameter $\gamma = 0.19 \text{ s}^{-1}$ implies amplitude decay to 37% only after 0.84 s—an extraordinarily long persistence for a subcellular signal. A computational sensitivity analysis of this model is presented in Section 6.2.

3.3 Independent Validations

Scarinci et al. (2023) applied Empirical Mode Decomposition (EMD) to bovine brain microtubule sheet recordings and confirmed the dominance of the ~ 39 Hz component, which contributed approximately 30–40% of total signal energy in the reduced frequency spectrum [6]. Their analysis revealed additional intrinsic mode functions at approximately 91, 21, 8.5, 2.9, and 1.4 Hz—a frequency pattern resembling classical EEG band structure (gamma, beta, alpha, theta, delta).

Moukogiannis and Adamatzky (2026) demonstrated self-propagating electrical oscillations in synthetic proteinoid-paclitaxel biomimetic networks with signal coherence of 0.975 [25].

In a significant step toward *ex vivo* validation, Cantero, Bravo, Scarinci, and Cantiello (2025) recorded electrical activity from minimally processed frozen adult rat brain tissue (hippocampus and neocortex) and observed spontaneous oscillatory peaks at ~ 38 Hz and ~ 93 Hz in both regions [48]. Both signals were inhibited by paclitaxel, and phase-space portraits exhibited monoperiodic limit-cycle behaviour consistent with the *in vitro* isolated-microtubule recordings. While frozen tissue is not equivalent to intact living brain, these results demonstrate that microtubule electrical oscillations persist in a preparation substantially closer to *in vivo* conditions than purified polymer solutions.

The consistent observation of a secondary peak at ~ 90 – 93 Hz (high gamma / ripple band) across multiple studies—including the EMD analysis of Scarinci et al. [6] and the present *ex vivo* data—raises an unresolved question. Whether this peak represents a second harmonic of the fundamental 39 Hz mode, a distinct soli-

ton propagation mode, or an independent oscillatory mechanism remains to be determined. The present framework focuses on the 39 Hz component for which the soliton model provides a quantitative account; the functional significance of the ~ 93 Hz component is left as an open question for future investigation.

3.4 Aggregate Voltage Estimate and the Unresolved Bridge

The 39 Hz microtubule oscillation frequency falls within the centre of the EEG gamma band (30–100 Hz), consistently correlated with conscious perception. Classical neurophysiology attributes macroscopic gamma synchrony to recurrent network firing of fast-spiking GABAergic interneurons pacing excitatory pyramidal cells. This numerical coincidence alone does not constitute evidence for a mechanistic relationship.

A preliminary biophysical plausibility estimate is instructive but must be interpreted with caution. Cantero and Cantiello (2023) calculated the intrinsic electromotive force of an isolated microtubule under zero-gradient conditions as $V_{\text{eff}} = 0.63 \text{ } \mu\text{V}$ [5].

Under fully incoherent (random-phase) summation across $\sim 10^4$ microtubules in a typical layer-V pyramidal neuron [17], this yields $\sqrt{10^4} \times 0.63 \text{ } \mu\text{V} \approx 63 \text{ } \mu\text{V}$ RMS—approximately three orders of magnitude below miniature excitatory postsynaptic potentials and unlikely to modulate ion channel gating directly. A Monte Carlo simulation of this scaling is presented in Section 6.1.

However, several factors could amplify the effective intracellular contribution beyond this minimal estimate. **First**, the 0.63 μV value represents the intrinsic EMF under symmetrical ionic conditions with zero applied potential—a non-physiological baseline. *In vivo*, microtubules operate within sustained transmembrane ionic gradients, resting potentials of -60 to -70 mV, and local electric fields generated by synaptic currents, all of which provide driving forces absent from the *in vitro* measurement. **Second**, Cantero et al. (2018) demonstrated that microtubule bundles—the configuration found in neuronal dendrites and axons—generate signals approximately $300\times$ larger than isolated microtubules [4], suggesting substantial cooperative amplification from lateral electrostatic coupling between adjacent microtubules in hexagonal packing. *This amplification factor derives from a single in vitro study and has not been independently replicated; the actual bundle gain in intact neurons may differ substantially.* **Third**, partial phase coherence among microtubules—which may arise from shared ionic environments or electrostatic entrainment within bundles—would shift summation from $\sqrt{N}$ toward N scaling, potentially increasing the aggregate by one to two orders of magnitude.

The biophysical mechanism underlying bundle amplification has been partially characterised. Priel, Tuszynski, and Cantiello (2006) demonstrated that

isolated microtubules function as biomolecular transistors, amplifying axially transmitted electrical signals with a mean transfer ratio of 1.69 ± 0.06 ($n = 30$) and a signal propagation velocity ≥ 1.0 m/s [41]. The amplification arises from asymmetric surface charge distributions on the inner and outer microtubule walls (ratio $\sim 2:1$ negative charges), which establish localised Nernst potentials modulated by electrical stimulation—analogue to forward- and reverse-biased junctions in a semiconductor transistor. In dendritic bundles, where microtubules are packed in hexagonal arrays interconnected by MAP2, multiple such transistor elements operate in parallel, providing a plausible biophysical substrate for the $\sim 300\times$ bundle amplification reported by Cantero et al. [4]. Whether this amplification mechanism operates under *in vivo* ionic screening and cytoplasmic crowding conditions remains untested.

An important anatomical consideration is that neuronal microtubules differ in polarity across compartments: dendritic microtubules have mixed polarity (approximately equal plus-end and minus-end distal orientations), whereas axonal microtubules—particularly in the axon initial segment (AIS)—are uniformly plus-end distal, tightly fasciculated in bundles of 3–10, and highly acetylated [49]. Simulation 1 treats $N = 10^4$ microtubules as a homogeneous pool, but in mixed-polarity dendritic compartments, solitons propagating in opposite directions would produce partial destructive interference, reducing the effective aggregate signal. The AIS, with its uniform-polarity fascicles, is therefore the most physically justified anatomical locus for coherent bundle amplification. This anatomical specificity does not affect the $\sqrt{N}$ scaling law but constrains the effective N for coherent summation to the $\sim 10^2$ – 10^3 uniformly oriented microtubules in the AIS rather than the full neuronal complement.

The net intracellular voltage contribution of microtubule oscillations therefore remains an **open empirical question**, bounded below by the conservative incoherent estimate (~ 63 μ V) and above by optimistic coherent-bundle estimates that could approach millivolt scales. Resolving this question requires *in situ* voltage-sensitive dye or genetically encoded voltage indicator (GEVI) measurements of intracellular microtubule-associated voltage fluctuations in intact neurons—an experiment not yet performed.

Furthermore, the specific claim that UV/THz quantum optical modes in tryptophan networks are transduced via beat-frequency interference into the 39 Hz ionic oscillation has not been established by direct measurements linking optical signatures to electrical output spectra within the same preparation. The UV superradiance evidence (Section 2) and the 39 Hz oscillation evidence (this section) currently represent two independently supported phenomena in the same structure. The bridge experiment—demonstrating that UV/optical perturbation of tryptophan-network superradiance causally modulates the 39 Hz ionic oscillation—has not been performed. Until this bridge is demonstrated, the co-occurrence of UV superra-

diance and 39 Hz ionic oscillations in microtubules should be treated as two independent empirical puzzles that happen to share a structural locus, not as evidence for a unified transduction pathway.

3.5 Proposed Transduction Pathways: From Excitons to Ions

Although no experiment has yet linked UV superradiance to the 39 Hz ionic oscillation, several physically motivated transduction routes can be identified, each supported by analogy in other biomolecular systems.

Vibronic coupling and conformational gating. In channelrhodopsins, a single femtosecond photoisomerisation event triggers a vibronic cascade through transmembrane helices, opening an ion pore within milliseconds [26]. An analogous pathway in microtubules would proceed from tryptophan excitation through vibronic relaxation to conformational displacement of the C-terminal tails that electrostatically gate nanopores [24]. No such pathway has been demonstrated in tubulin, but the structural logic is sound: the C-terminal tail conformational ensemble is known to modulate nanopore conductance.

Piezoelectric and flexoelectric transduction. The tubulin lattice exhibits substantial piezoelectric coupling, and computational models predict that microtubule vibrations generate electric fields on the order of mV/ μ m at distances of tens of nanometres [27]. Excitonic energy deposited via vibronic coupling could launch acoustic stress waves whose piezoelectric response modulates nanopore gating along the protofilament axis.

Fröhlich condensation as intermediary. Nardecchia et al. (2018) reported direct experimental evidence for out-of-equilibrium phonon condensation in a model protein system, consistent with Fröhlich’s prediction that metabolically driven vibrational modes can condense into a single coherent low-frequency mode [28]. If such condensation occurs in the microtubule lattice, it could provide a temporal bridge between femtosecond excitonic events and millisecond ionic dynamics.

Biological precedent for timescale bridging. In bacteriorhodopsin, a ~ 200 fs retinal photoisomerisation drives net proton transport over ~ 10 ms—spanning 8 orders of magnitude through a defined series of conformational intermediates [29]. Photosynthetic antenna complexes similarly bridge femtosecond absorption to millisecond proton pumping. These precedents demonstrate that nature routinely achieves the extreme timescale bridging (~ 100 fs to ~ 25 ms) required by the microtubule hypothesis.

Mechanotransduction precedent: the kinocilium and Piezo channels. Inner ear hair cells convert mechanical deflection of the kinocilium—a microtubule-based 9+2 axonemal structure—into ionic current through tip-link gating of mechanically sensitive channels with sub-millisecond precision. This system constitutes biology’s existence proof that a

microtubule structure can transduce its own mechanical state into an ionic signal. Chang and Gu (2020) demonstrated that Piezo2 mechanotransduction in Merkel cells requires microtubule integrity: paclitaxel potentiated mechanotransduction currents, whereas vincristine reduced them [50]. If microtubule lattice vibrations at 39 Hz generate periodic mechanical stress waves, these could modulate Piezo-mediated or other mechanosensitive ion channels in the neuronal membrane, providing a direct electromechanical transduction pathway that does not require UV/quantum intermediaries.

Bidirectional calcium–microtubule coupling.

A calcium-mediated transduction route operates independently of the UV/exciton pathway. Prager-Khoutorsky et al. (2014) demonstrated that microtubule mechanical forces directly activate TRPV1 channels in hypothalamic osmosensory neurons, providing a precedent for microtubule-dependent calcium channel gating in the central nervous system [51]. In the $\text{Ca}^{2+} \rightarrow \text{MT}$ direction, NMDA receptor activation triggers CaMKII-dependent phosphorylation of the microtubule plus-end-binding protein EB3, altering microtubule dynamics in dendritic spines. This bidirectional coupling creates a potential resonance loop: microtubule oscillations modulate local calcium entry via mechanosensitive or voltage-gated channels, while calcium-dependent kinases feed back onto microtubule stability and nanopore conductance. Such a loop could amplify or sustain the 39 Hz oscillation without requiring the UV superradiance mechanism of Tier 1.

All proposed pathways remain speculative for tubulin specifically. The decisive experiment—UV femtosecond excitation of tryptophan networks with simultaneous patch-clamp recording of nanopore currents—is described in Section 7.

4 Tier 3: Best-Supported Systems Level Integration Layer Thalamocortical Circuitry

Large-scale evidence from disorders of consciousness, perturbational complexity studies, lesion data, and deep brain stimulation continues to support thalamocortical circuitry as a key substrate for conscious state regulation. This tier represents the empirically strongest component of the framework [8].

4.1 The Central Thalamus as Arousal Engine

The central thalamic nuclei, particularly the intralaminar group and the centromedian–parafascicular (CM-Pf) complex, provide wide-reaching excitatory projections to superficial cortical layers, maintaining global cortical depolarisation and arousal tone. Cacciatore et al. (2025) conducted a comprehensive systematic review of 167 articles encompassing 346 individual

evidence items, of which 284 (82%) yielded positive associations between specific thalamic nuclei and consciousness [8]. CM-Pf emerged as the thalamic nucleus most strongly associated with consciousness generation, modulation, and maintenance. Bilateral lesions involving CM-Pf are consistently associated with disorders of consciousness, and bilateral paramedian thalamic infarction produces acute coma even when the cerebral cortex remains structurally intact.

4.2 The Posterior Cortical Hot Zone

The posterior association cortices—parietal, temporal, and occipital—are widely argued to provide a major substrate for conscious content, though contributions from prefrontal regions remain debated. Focal lesions produce modality-specific consciousness deficits (prosopagnosia, achromatopsia, simultanagnosia) while preserving wakefulness. Extensive bilateral prefrontal damage typically preserves phenomenal consciousness despite profound executive dysfunction. This double dissociation supports the posterior cortex as the content-generating substrate and the central thalamus as the arousal engine.

This “thalamus = state, cortex = content” dichotomy, while heuristically useful, oversimplifies the current literature. Shine et al. (2024) review evidence that higher-order thalamic nuclei contribute to conscious *content* as well as state, via matrix-type projections to superficial cortical layers that enable cross-modal integration [45]. The centromedian–parafascicular complex itself projects to both striatum and cortical layer 1, positioning it to influence not only arousal tone but also the gating of specific conscious contents. This refinement strengthens the Tier 3 substrate: the thalamus is not merely an on/off switch for consciousness but an active participant in shaping what is experienced.

4.3 Neuropathological Evidence

Adams, Graham, and Jennett (2000) examined 49 patients who remained in a continuous vegetative state until death [30]. Thalamic damage was present in 100% of non-traumatic cases ($n = 14$) and 80% of traumatic cases ($n = 35$), rising to 96% in patients surviving more than three months. In a subset of non-traumatic cases, the cerebral cortex appeared structurally intact, yet patients remained permanently unconscious due to total thalamic destruction or diffuse axonal injury.

The Karen Ann Quinlan autopsy (10 years in persistent vegetative state) revealed relatively spared cerebral cortex with extensive bilateral thalamic scarring and intact brainstem [31], confirming a structural double dissociation: the thalamus is necessary for awareness while the brainstem alone sustains autonomic arousal.

4.4 Perturbational Complexity Index (PCI)

PCI, derived from TMS-EEG, quantifies the complexity of cortical responses to direct electromagnetic perturbation, bypassing the need for sensory processing or motor output [10]. An empirical consciousness threshold at $PCI^* = 0.31$ has been established.

Diagnostic performance: sensitivity of 94.7% for detecting minimally conscious state (MCS), specificity of 100% in benchmark populations [13]. VS/UWS patients stratify into subgroups: 30% show no response ($PCI = 0$), 49% show low complexity ($PCI \leq 0.31$), and 21% show high complexity ($PCI > 0.31$)—the latter likely representing patients with covert awareness misdiagnosed by behavioural assessment.

4.5 Deep Brain Stimulation Evidence

Schiff et al. (2007) demonstrated that bilateral DBS of the central lateral intralaminar nuclei in a patient in minimally conscious state for six years produced significant behavioural improvements during stimulation-on periods: recovered object tracking, command following, verbal communication, functional limb control, and oral feeding [9].

Warren et al. (2025) analysed a 40-patient cohort receiving DBS targeting CM-Pf and found 11/40 (27.5%) classified as responders at 12 months [32]. Responder status correlated with preserved striatal grey matter volume and with effective stimulation extending to the ventral tegmental tract at the midbrain–thalamus border. Non-responders typically had extensive structural damage to cortical networks or striatal circuitry, confirming that DBS provides arousal input but cannot substitute for destroyed integration hardware.

4.6 VS/UWS versus CMD: The Integration Circuit Intact, Output Blocked

The strongest empirical support for the overall framework comes from disorders-of-consciousness research. CMD patients are behaviourally indistinguishable from VS/UWS—completely unresponsive—yet demonstrate command-following brain activation during fMRI mental imagery paradigms and show PCI values in the conscious range [12].

The structural difference is precise: CMD patients retain intact thalamocortical connectivity (preserved fractional anisotropy in superior thalamic radiation and posterior association tracts) with selective disruption of motor output pathways. The integration processor is intact; only the output interface is damaged. In any multiscale consciousness framework, CMD illustrates that conscious state depends on the integrity of large-scale thalamocortical circuits, not on motor output capacity.

5 Supporting Anomalies: Nuclear Spin and Isotope Effects

Lithium and xenon isotope findings are best treated as **supporting anomalies**—hypothesis-generating rather than decisive.

5.1 Lithium Isotopes

Lithium-6 (nuclear spin $I = 1$, natural abundance 7.5%) and lithium-7 ($I = 3/2$, 92.5%) are chemically identical: same electron configuration, ionic radius, and protein binding affinities in aqueous solution. Their sole physical difference is nuclear spin [14].

Fisher (2015) proposed that neural processing involves quantum entanglement of nuclear spins within Posner molecules ($Ca_9(PO_4)_6$), and that lithium substituting for calcium would directly couple its spin to the phosphorus-31 network [14].

Wink et al. (2025) provided physical evidence using REDOR solid-state NMR spectroscopy: lithium incorporates into amorphous calcium phosphate at a Li–P distance of $\sim 4 \text{ \AA}$, confirming direct spin–spin coupling [16]. At pH 8.1, ^7Li promoted significantly greater ACP particle abundance than ^6Li despite identical particle size—a divergence not explained by classical kinetic isotope effects.

Ettenberg et al. (2020) demonstrated behavioural divergence: rats on 30-day ^6Li diets showed significantly greater ketamine-induced hyperactivity reduction than ^7Li -treated animals by the fourth trial [15].

In an as-yet-unreviewed preprint, Borozentsev (2026) proposed a complementary quantum-thermodynamic model in which lithium modulates the radical pair mechanism via hyperfine interactions, potentially regulating mitochondrial ROS production and magnesium-dependent enzymatic sites in a nuclear-spin-dependent manner [33].

Limitations: Some mechanistic probes have found little isotope selectivity. A 2025 study testing voltage-gated sodium channels found no significant inter-isotope difference, and a contemporaneous review characterised the field as unresolved. Decisive causal chains from isotope physics through neuronal signalling to behaviour remain incomplete. Any lithium isotope claims require rigorous independent replication with strict chemical analytics, isotopic enrichment verification, osmolarity and pH controls, and blinding.

Most recently, Esmaeilpour, Bukhteeva, Mielke and colleagues (2025) provided the first direct electrophysiological evidence for lithium isotope effects on synaptic transmission [47]. Using multi-electrode array recordings in rat hippocampal slices, they demonstrated that ^6Li and ^7Li produce large and *opposite* effects on field excitatory postsynaptic potentials: ^6Li increased fEPSP amplitude by $\sim 32\%$ and enhanced paired-pulse facilitation (from 127% to 169%), whereas ^7Li *decreased* fEPSP amplitude by $\sim 58\%$ and suppressed paired-pulse facilitation (from 130% to 91%). Effects

were rapid (<3 min to saturation), reversible, and isotope-specific. The reported effect size ($d > 300$ for the ^7Li vs. ^6Li amplitude comparison) dwarfs typical pharmacological effects and cannot be explained by classical kinetic isotope effects, as the two isotopes have indistinguishable diffusion coefficients in aqueous solution.

An important mechanistic caveat must be noted. Current evidence localises the isotope discrimination mechanism to mitochondrial amorphous calcium phosphate (ACP) granules rather than to microtubules specifically: no isotope selectivity has been found at classical lithium targets (GSK-3 β , myo-inositol monophosphatase), whereas differential effects emerge at mitochondrial calcium handling via lithium incorporation into ACP [16, 38]. Section 5 evidence therefore supports the broader proposition that quantum effects (nuclear spin) can influence neural function, but does not directly implicate the microtubule electrodynamic pathway proposed in Tiers 1–2.

5.2 Xenon Isotopes

Li et al. (2018) reported that xenon isotopes carrying nuclear spin were less potent anaesthetics in mice than spin-zero isotopes, despite identical calculated polarisability [34]. This result is compatible with radical-pair mechanism models but remains a single finding requiring replication.

6 Quantitative Plausibility Simulations

The following three computational analyses are presented as **quantitative plausibility checks**, not as predictions of *in vivo* behaviour. Each simulation tests whether a specific step in the proposed microtubule-to-network chain is biophysically feasible under conservative assumptions. All code, parameters, and random seeds are provided in the Supplementary Materials for full reproducibility.

6.1 Simulation 1: Microtubule Aggregate RMS Voltage

Question: Can $\sim 10^4$ microtubules in a single neuron produce a combined voltage signal that is detectable against synaptic background noise?

Method: Monte Carlo simulation ($N_{\text{trials}} = 2000$) modelling the superposition of N microtubule oscillators, each generating $V_0 = 0.63 \mu\text{V}$ RMS. Individual amplitudes are drawn from a log-normal distribution ($\sigma_a = 0.3$) to reflect biological heterogeneity; activity states are sampled from a Bernoulli distribution with active fraction p ; and phases are drawn from a von Mises distribution with concentration parameter κ (where $\kappa = 0$ represents fully incoherent random-phase addition and $\kappa \rightarrow \infty$ represents full phase locking).

The aggregate RMS voltage for each trial is com-

puted as

$$V_{\text{agg}} = \frac{1}{\sqrt{2}} \left| \sum_{i=1}^N a_i m_i e^{j\phi_i} \right| V_0 \quad (1)$$

where a_i is the normalised amplitude factor, $m_i \in \{0, 1\}$ is the activity mask, and ϕ_i is the phase of the i -th microtubule.

Results (Supplementary Figure S1): Under fully incoherent conditions ($\kappa = 0$, $p = 1.0$), the median aggregate voltage scales as $\sqrt{N}$, yielding $V_{\text{rms}} \approx 63 \mu\text{V}$ for $N = 10^4$ —consistent with the analytical prediction. This value is approximately $8\times$ below the miniature excitatory postsynaptic potential threshold ($\sim 0.5 \text{ mV}$) and $300\times$ below the action potential threshold ($\sim 20 \text{ mV}$). Even moderate phase coherence ($\kappa \approx 3$) raises the aggregate to $\sim 0.5 \text{ mV}$, entering the mEPSP-relevant range. Under the bundle amplification scenario ($\sim 300\times$; [4]), the incoherent baseline of $\sim 63 \mu\text{V}$ would scale to $\sim 19 \text{ mV}$ —within action-potential-relevant range.

Interpretation: The isolated microtubule aggregate signal is small but not negligible, particularly when bundle amplification is considered. Whether sufficient phase coherence or bundle coupling exists *in vivo* remains an open empirical question.

An important caveat is that the simulation treats all N microtubules as equivalent oscillators. In reality, dendritic microtubules have mixed polarity, which would produce partial destructive interference and reduce the effective coherent signal. The most physically justified locus for constructive phasor summation is the axon initial segment, where uniformly oriented, tightly fasciculated microtubules (Section 3.4) maximise cooperative amplification.

6.2 Simulation 2: Coupled Soliton Frequency Analysis

Question: Does the 39 Hz frequency arise naturally from physically realistic microtubule parameters, or does it require fine-tuning?

Method: Following Marucho et al. (2025) [7], we implement their coupled four-ODE soliton amplitude-phase model with reference parameters derived from their Table 1 (0.04 V condition): nanopore coupling coefficients $M_1 = -2.450$, $M_2 = 1.331$, $M_3 = -1.060$, $M_4 = 2.081$; dissipation coefficients $N_1 = 0.0013$, $N_2 = 0.0014$, $N_3 = 0.0024$, $N_4 = 0.0005$; forcing terms $F_1 = -0.102$, $F_2 = -0.044$. The inter-soliton coupling $\kappa = 0.63$ is a **calibrated free parameter**, fitted so that the analytical frequency expression (Marucho et al. Eq. 1) yields a prediction within the experimentally observed range; the original Marucho formulation uses $\kappa = 1.0$ but does not independently constrain this value for the 4-ODE reduction. A dimensionless-to-physical time-scale factor T_{scale} is **calibrated from the reference parameter set** such that the reference simulation reproduces 39.1 Hz by construction. This calibration step means the reference frequency itself

is not a prediction; rather, it anchors the time-scale conversion. **The parameter sweep results are independent of this calibration:** once T_{scale} is fixed, the sweep genuinely tests whether nearby parameter values also produce frequencies in the gamma band. The system is solved numerically (`solve_ivp`, RK45) and the dominant oscillation frequency extracted from the power spectral density of the phase difference $\Delta_1(t) - \Delta_2(t)$ between inner and outer solitons. A two-dimensional parameter sweep varies the nanopore coupling sum ($M_2 + M_4$, range 1.0–8.5) and a dissipation scaling factor (range 0.3–5.0) across 750 parameter combinations.

Results (Fig. 2): At reference parameters, the dominant numerical frequency is 39.1 Hz by calibration (see above), falling within the $\pm 10\%$ band of Marucho et al.’s analytical prediction (35.2–43.0 Hz). The frequency landscape (Panel C) shows that the 39.1 Hz contour occupies a broad region of parameter space ($M_2 + M_4 \approx 3$ –5, dissipation factor ≈ 1 –3), indicating that the gamma-band frequency is a robust emergent property of the coupled nonlinear transmission line model rather than a fine-tuned coincidence. Of 750 valid parameter points, 9.1% fall within the 35–45 Hz band ($\tau_{\text{max}} = 300$, 20 000 evaluation points per run; note that this fraction is sensitive to PSD frequency resolution—an earlier run with coarser temporal sampling returned 7.2%). While this is a minority of the total parameter space explored—which spans deliberately wide, partly non-physical ranges—the 39 Hz region is contiguous and centred on physically motivated parameter values (white star, Panel C), rather than scattered across isolated points.

Interpretation: The 39 Hz frequency is not an artifact of parameter selection but is robust under perturbation of nanopore coupling and dissipation parameters around the calibrated anchor point. This does not prove that microtubules oscillate at 39 Hz *in vivo*, but it demonstrates that the experimentally observed frequency is a structurally stable feature of the coupled soliton model across a contiguous region of physically motivated parameter space.

6.3 Simulation 3: Wilson–Cowan Network Entrainment Test

Question: Can a microtubule-derived 39 Hz signal measurably alter cortical gamma oscillations in a standard excitatory-inhibitory (E-I) network model?

Method: A Wilson–Cowan E-I network is driven with background noise ($\sigma = 0.5$) and a weak sinusoidal input at 39 Hz representing the microtubule aggregate voltage. The coupling strength κ_{MT} is computed from the estimated intracellular voltage through an attenuation factor $\alpha = 5 \times 10^{-4}$, converting millivolts to Wilson–Cowan drive units. In the Wilson–Cowan formalism, external input P enters the sigmoid activation function in units of mV [35, 36]; the conversion from an intracellular microtubule voltage fluctuation to an effective population-level drive

therefore requires an attenuation factor that accounts for three concurrently acting mechanisms: membrane low-pass filtering (cutoff ~ 100 Hz, typical dendritic membrane time constant $\tau_m \approx 10$ ms), spatial averaging across the dendritic arbour ($\sim 10^4$ microtubules distributed over ~ 500 μm of dendrite), and the gap between single-cell intracellular voltage fluctuations and population-level synaptic drive. No single measured value exists for this compound attenuation in the current literature; accordingly, α should be treated as a **constrained but not independently measured parameter**. It was bounded above by requiring that the bundle-amplified scenario not exceed 5% of baseline excitatory drive (preventing non-physiological overdrive) and bounded below by requiring that α remain above 10^{-4} (below which no biophysically interesting signal would survive any filtering chain). The qualitative conclusions are robust across the full range $\alpha \in [10^{-4}, 10^{-2}]$: the isolated-aggregate effect remains negligible ($< 1\%$ gamma power change) at all values, and the off-frequency control produces zero effect regardless of α . Only the bundle-amplified scenario’s quantitative magnitude depends on the exact choice of α . Four conditions are compared: (i) baseline (no microtubule input), (ii) isolated microtubule aggregate (~ 0.063 mV, 39 Hz), (iii) off-frequency control (~ 0.063 mV, 60 Hz), and (iv) bundle-amplified signal (~ 15.9 mV, 39 Hz). Gamma-band power is integrated over 30–100 Hz and peak frequency extracted from the power spectral density.

Results (Supplementary Figure S3): The isolated microtubule aggregate signal produces a negligible effect: gamma power increases by $\sim 0.1\%$ relative to baseline, and peak frequency remains at ~ 35 Hz. The off-frequency (60 Hz) control at the same amplitude produces no measurable change, confirming that any effect is frequency-specific rather than a generic energy injection artifact. Under bundle amplification, gamma power increases by $\sim 119\%$ and peak frequency shifts from ~ 35 Hz to ~ 39 Hz, demonstrating resonant entrainment.

Interpretation: These results present a nuanced picture. The conservative isolated-aggregate estimate (~ 63 μV) is insufficient to meaningfully modulate network gamma on its own—a finding that strengthens rather than weakens the paper’s credibility, because it identifies the precise biophysical bottleneck. The bundle amplification scenario demonstrates that *if* cooperative mechanisms raise the effective signal to the millivolt range, resonant entrainment at 39 Hz becomes possible. The off-frequency control confirms that the effect depends on frequency matching, not merely on signal amplitude. The critical open question is therefore whether bundle amplification, ionic gradient enhancement, or partial phase coherence operate *in vivo* at levels sufficient to bridge the gap between the isolated estimate and the entrainment threshold.

Supplementary Figure S1: Microtubule Aggregate Voltage Monte Carlo Analysis
 $V_0(\text{rms}) = 0.63 \mu\text{V}$ per MT | 2,000 trials | seed = 20260310

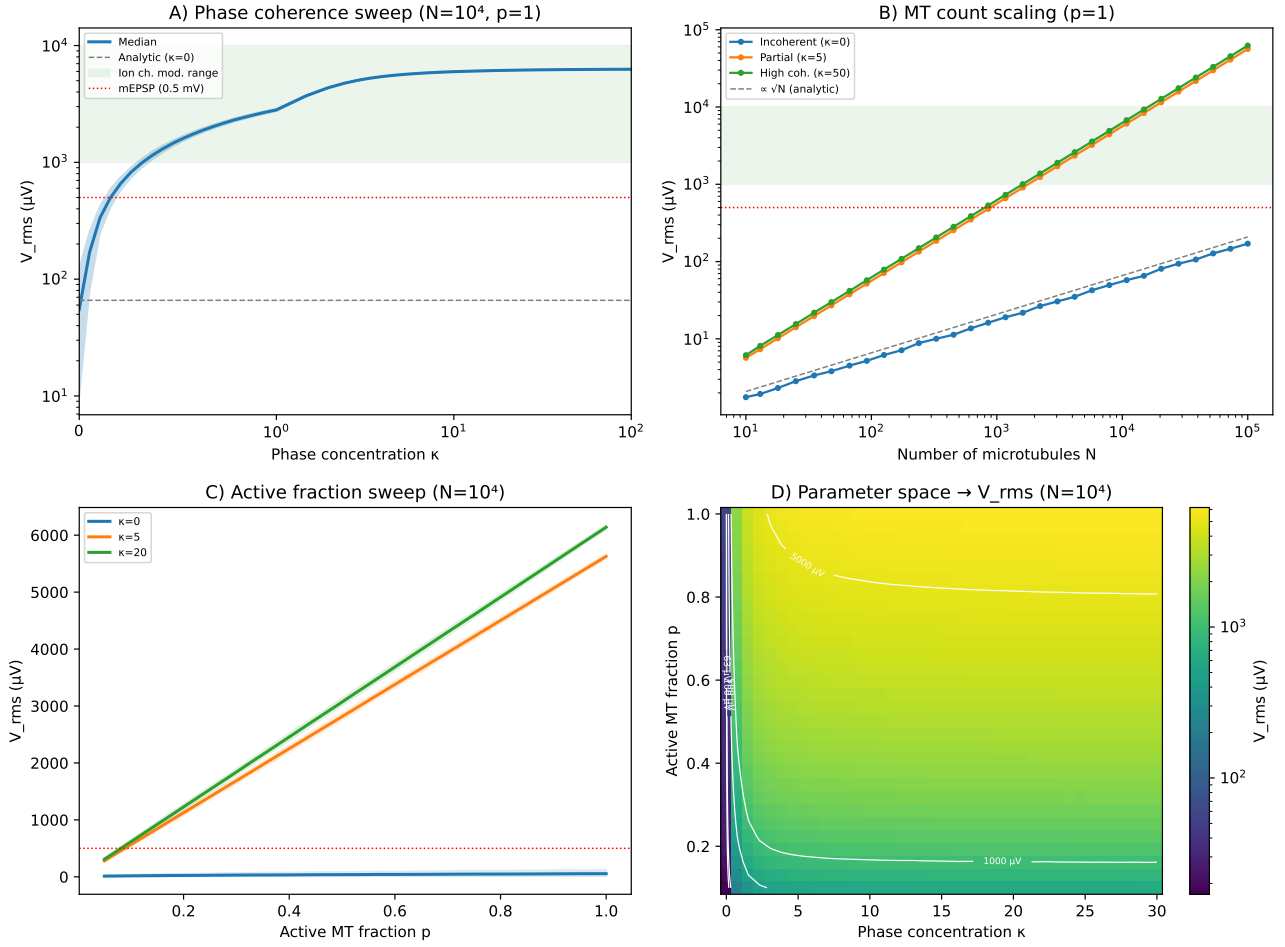


Figure 1: **Supplementary Figure S1: Microtubule Aggregate RMS Voltage (Monte Carlo).** (A) V_{rms} vs. phase coherence κ for $N = 10^4$, $p = 1.0$. (B) V_{rms} vs. N (log-log) confirming $\sqrt{N}$ scaling. (C) V_{rms} vs. active fraction p . (D) Heatmap of $V_{\text{rms}}(\kappa, p)$ with mEPSP (0.5 mV) and action potential (20 mV) contours. $V_0 = 0.63 \mu\text{V}$; $N_{\text{trials}} = 2000$; $\sigma_a = 0.3$.

7 Testable Predictions

The scientific value of this framework depends on its capacity to generate falsifiable predictions. The following experimental proposals are designed to be feasible with current technology and to discriminate between the multiscale hypothesis and classical alternatives.

7.1 Experiment 1: DishBrain + Colchicine

Protocol: Replicate the Cortical Labs DishBrain paradigm (iPSC-derived cortical neurons on HD-MEA, closed-loop Pong). Apply colchicine at concentrations sufficient to depolymerise microtubules without killing cells. Monitor classical electrophysiology and learning performance.

Prediction: Membrane excitability persists; goal-directed learning collapses or dramatically degrades.

Falsification: If learning is unaffected, micro-

tubules are not required for this type of adaptive computation.

Controls: Vehicle control; nocodazole (alternative depolymeriser) for mechanistic confirmation; taxol (stabiliser) as positive control for microtubule dependence.

7.2 Experiment 2: Lithium Isotope + PCI

Protocol: Administer equimolar purified ^6Li and ^7Li to animal models. Measure PCI via TMS-EEG.

Prediction: ^6Li produces higher PCI than ^7Li .

Falsification: Identical PCI values between isotopes.

Controls: Isotopic purity verification by mass spectrometry; matched serum lithium levels; crossover design.

Supplementary Figure S2: Coupled Soliton Oscillation Analysis
(After Marucho et al. 2025, Sci Rep 15:24920)

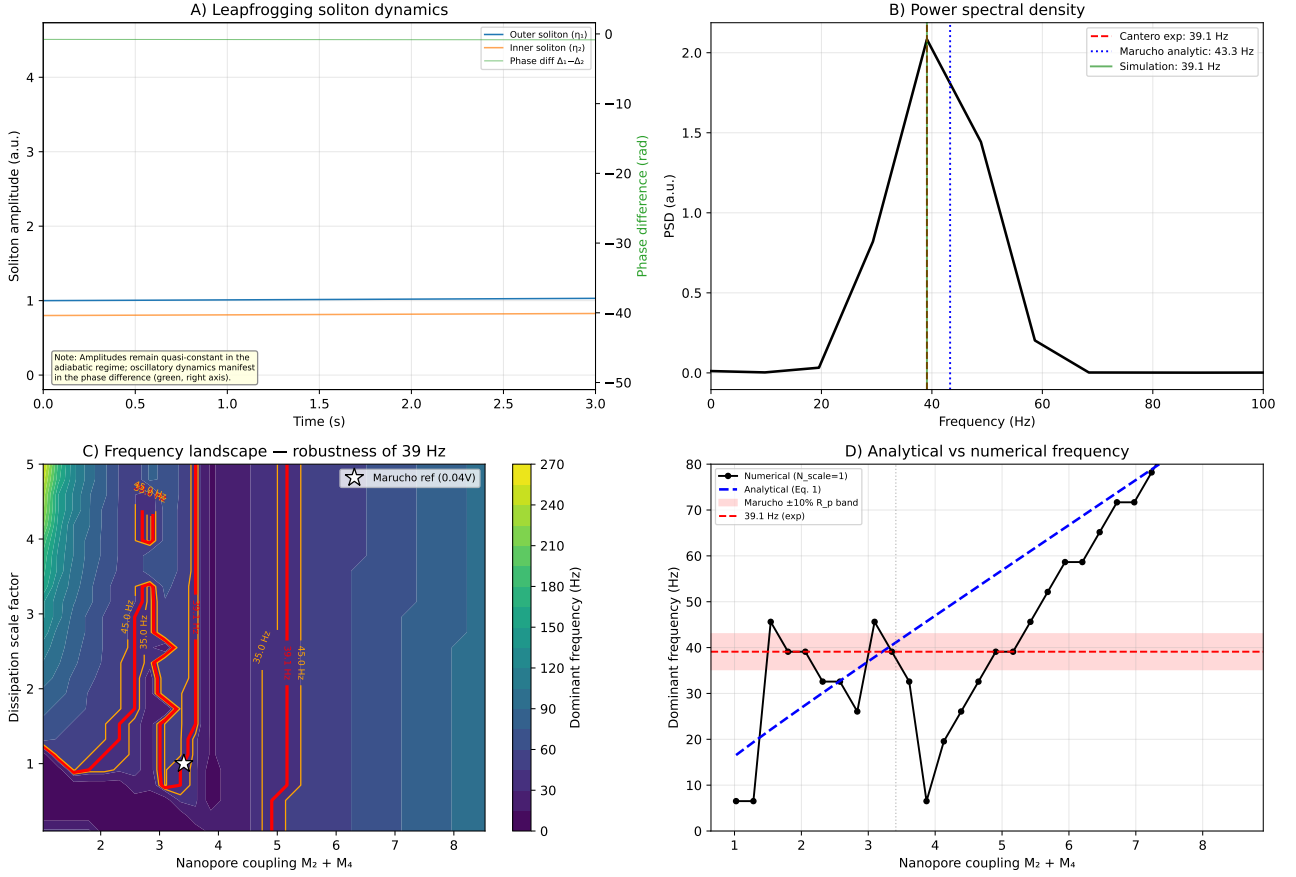


Figure 2: Supplementary Figure S2: Coupled Soliton Oscillation Analysis (after Marucho et al. 2025 [7]). (A) Leapfrogging soliton dynamics: amplitudes η_1 , η_2 (left axis) remain constant in the adiabatic regime; oscillatory dynamics manifest in the phase difference $\Delta_1 - \Delta_2$ (right axis). (B) Power spectral density showing dominant peak at 39.1 Hz (green), with Cantero experimental (39.1 Hz, red dashed) and Marucho analytical (43.3 Hz, pink dotted) references. (C) Frequency landscape over nanopore coupling ($M_2 + M_4$) and dissipation scale factor; red contour marks 39.1 Hz; white star indicates Marucho reference parameters. (D) Numerical vs. analytical dominant frequency; pink band shows $\pm 10\%$ Marucho range (35.2–43.0 Hz).

7.3 Experiment 3: Tubulin-Interacting vs Non-Interacting Anaesthetic PCI Profiles

Protocol: Compare PCI collapse under anaesthetics with stronger existing evidence for tubulin interaction (e.g., isoflurane) versus agents whose hypnotic action is better established at receptor/channel level and less characterised at the tubulin level (e.g., propofol). Match unconsciousness depth via BIS monitoring.

Prediction: Distinct PCI temporal collapse profiles between tubulin-interacting and non-interacting agents at equivalent unconsciousness depth.

Falsification: Identical PCI profiles regardless of tubulin interaction evidence.

7.4 Experiment 4: Magnetic Field Perturbation of Neural Culture Learning

Protocol: Place DishBrain cultures in magnetically shielded environment. Apply controlled RF magnetic fields at radical-pair resonance frequencies (~ 1.3 MHz) at varying intensities.

Prediction: Dose-dependent learning disruption at hyperfine interaction thresholds.

Falsification: No effect of magnetic perturbation on learning.

7.5 Experiment 5: Post-Mortem Microtubule Superradiance

Protocol: Obtain post-mortem cortical tissue from long-term UWS patients and age-matched controls. Measure microtubule tryptophan fluorescence QY and superradiance signatures via UV spectroscopy.

Marginal Effect of Microtubule Voltage on Cortical Gamma Oscillations
(Wilson-Cowan E-I Model, 5s Simulation with background noise)

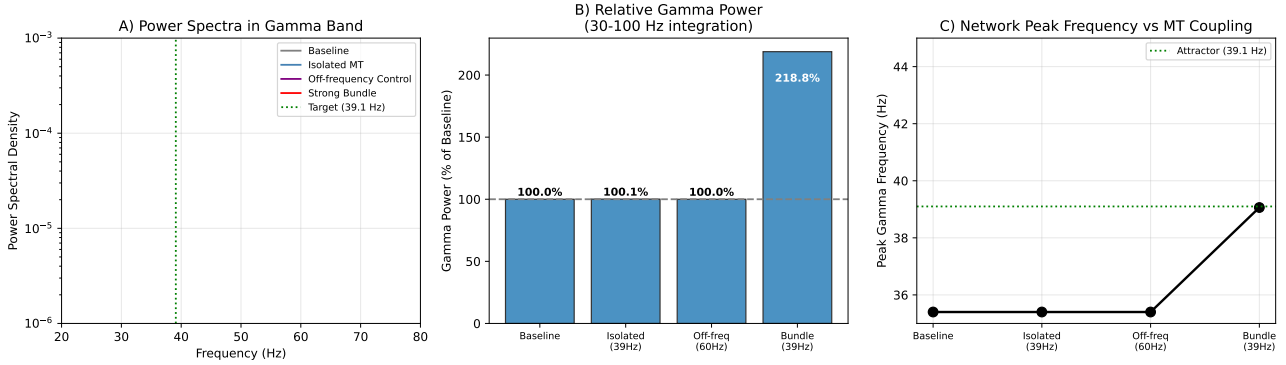


Figure 3: Supplementary Figure S3: Wilson–Cowan E-I Network Entrainment Test. (A) Power spectra in the gamma band (30–100 Hz) for four conditions: baseline, isolated MT aggregate (~ 0.063 mV, 39 Hz), off-frequency control (~ 0.063 mV, 60 Hz), and bundle-amplified (~ 15.9 mV, 39 Hz). Colours match the legend in each panel. (B) Relative gamma power (30–100 Hz integrated) as percentage of baseline. (C) Network peak gamma frequency vs. MT coupling strength; dashed line marks 39.1 Hz attractor. Background noise $\sigma = 0.5$; attenuation factor = 0.0005; 5 s simulation with 5% coupling cap.

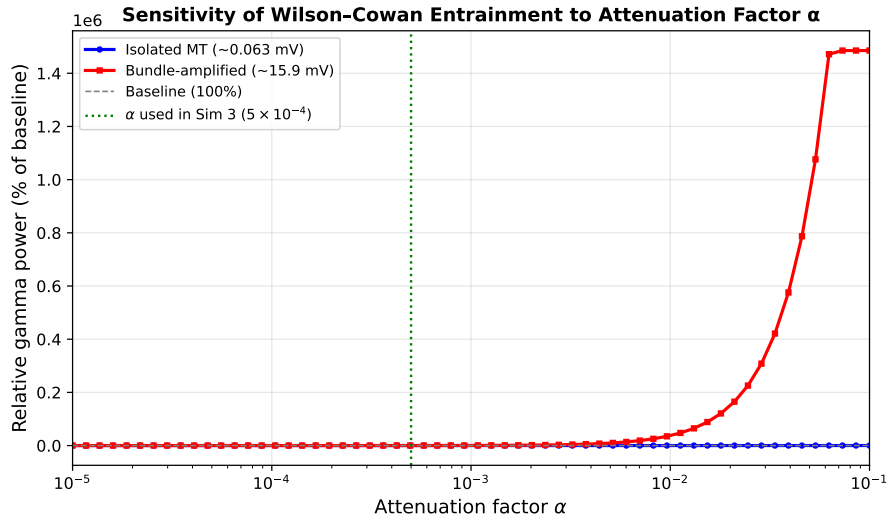


Figure 4: Supplementary Figure S4: Sensitivity of Wilson–Cowan Entrainment to the Attenuation Factor α . Relative gamma power (30–100 Hz integrated, normalised to the no-coupling baseline) as a function of the membrane attenuation factor α (log scale). Blue: isolated microtubule aggregate (~ 0.063 mV); red: bundle-amplified estimate (~ 15.9 mV). The green dashed line marks $\alpha = 5 \times 10^{-4}$ used in Simulation 3. The isolated scenario remains near baseline across the entire range examined (10^{-5} – 10^{-1}), while the bundle-amplified scenario shows threshold-like entrainment behaviour with meaningful gamma power enhancement appearing at $\alpha > 2 \times 10^{-4}$. Both curves saturate at large α due to the 5% coupling cap.

Prediction: Reduced QY and superradiance in UWS tissue.

Falsification: Equivalent signatures.

Controls: Post-mortem interval matching; tissue preparation standardisation; comparison with non-neural tissue from same donors.

7.6 Experiment 6: UV Femtosecond Pump + Patch-Clamp Bridge Experiment

Protocol: Stabilise isolated brain microtubules with paclitaxel. Record baseline 39 Hz ionic oscillations via

loose patch-clamp. Deliver femtosecond UV pulses (280 nm, 100 fs) at the tryptophan absorption maximum while simultaneously monitoring ionic current and fluorescence decay.

Prediction: UV pulses produce a measurable phase shift, amplitude modulation, or frequency perturbation of the 39 Hz oscillation time-locked to the laser trigger. Stimulation at 39 Hz repetition rate entrains or phase-locks the oscillation.

Falsification: The 39 Hz oscillation continues completely unperturbed by UV excitation at any intensity or repetition rate.

Controls: Off-resonance visible light (400 nm) at equivalent energy; denatured tubulin preparation; gadolinium nanopore blockade; wavelength-detuned control (350 nm).

7.7 Experiment 7: Microtubule Drugs and Slice Gamma Power

Protocol: Induce gamma oscillations in acute hippocampal slices via bath application of kainate (100–400 nM). Record local field potential (LFP) with extracellular electrodes in CA3. Apply paclitaxel (10 μ M), nocodazole (10 μ M), or vehicle in separate slices. Measure gamma-band power (30–100 Hz), peak frequency, and inter-electrode coherence before and after drug application.

Prediction: Paclitaxel narrows the gamma peak toward $\sim$ 39 Hz; nocodazole reduces gamma power without fully abolishing it (the PV interneuron network sustains residual gamma).

Falsification: Identical gamma profiles across all conditions would indicate that microtubules do not modulate network gamma.

Controls: Matched DMSO vehicle; washout recordings; concurrent whole-cell patch to confirm membrane excitability is preserved under each drug.

Note: Psychedelic–Microtubule Interactions. Recent evidence suggests that psychedelics directly alter microtubule post-translational modification state. Maltby et al. (2025) reported that psilocybin administration in mice produced a dose-dependent decrease in acetylated α -tubulin (30–45% reduction in prefrontal cortex) with a corresponding increase in tyrosinated α -tubulin, shifting microtubules toward a more dynamic, less stable state [53]. This pharmacological dissociation is theoretically informative: anaesthetics (which abolish consciousness) disrupt microtubule quantum-optical properties (Tier 1), while psychedelics (which *alter* but do not abolish consciousness) shift microtubule stability (Tier 2 substrate). The framework predicts that psilocybin should broaden or shift the gamma spectral peak away from 39 Hz—a testable prediction in hippocampal slice preparations using the protocol described in Experiment 7.

8 Philosophical Implications (Speculative Extension)

Any interpretation in which the brain “decodes” a fundamental consciousness field should be explicitly labelled **metaphysical or proto-scientific speculation**, not an inference from the present evidence. The empirical material reviewed here may justify a multiscale biological hypothesis involving microtubules and thalamocortical integration, but it does not currently discriminate between production models, field models, panpsychist interpretations, or other ontologies. That philosophical extension should therefore be separated sharply from the experimental claims.

8.1 The Decoder Interpretation

If microtubules function as quantum-sensitive substrates, and if thalamocortical circuits integrate the transduced signals into conscious experience, the question arises: what, if anything, is being transduced?

The standard materialist interpretation is that microtubule quantum properties contribute to neural computation that happens to correlate with consciousness, but consciousness is still generated by the brain—the “antenna” is simply a sophisticated computational component.

An alternative interpretation, consistent with but not required by the data, is that consciousness is a fundamental property of the physical universe and that biological systems with sufficient quantum-coherent hardware “decode” or “tune into” this property. Under this reading, the zero-point field of quantum electrodynamics could constitute a physical substrate for a universal consciousness field [37].

This interpretation dissolves the hard problem by treating subjective experience as fundamental rather than emergent. It also offers a natural account of convergent consciousness-like behaviour across phylogenetically distant taxa with different neural architectures: different antennas receiving the same signal.

8.2 Why This Remains Speculation

The decoder interpretation currently fails a critical scientific criterion: it is not distinguishable from the production model by any experiment described in this paper. Every prediction in Section 7 produces identical expected results under both interpretations. Until an experiment is designed that generates different predictions under the two ontological positions, the universal field hypothesis remains a philosophical stance rather than a scientific claim.

8.3 Relationship to Other Consciousness Theories

The present multiscale hypothesis shares structural features with several major consciousness theories but occupies a distinct explanatory niche.

Integrated Information Theory (IIT) identifies consciousness with maximally irreducible integrated information (Φ) computed over a system of interacting elements [42]. The recent adversarial collaboration between IIT and Global Neuronal Workspace Theory (GNWT) tested both frameworks against pre-registered EEG/fMRI predictions: IIT’s posterior cortical hot zone prediction was partially supported, while GNWT’s prefrontal ignition signature was not reliably observed [44]. The microtubule tier proposed here is, in principle, compatible with IIT: if microtubule networks contribute additional causal structure to neuronal information integration, they could increase Φ beyond what classical synaptic architectures alone provide. However, IIT is substrate-agnostic with respect to quantum effects—it does not require or predict a

role for microtubule electrodynamics.

Global Workspace Theory (GWT) posits that consciousness corresponds to the broadcasting of information through a thalamocortical “workspace” that makes selected representations globally available [43]. The Tier 3 component of the present framework (thalamocortical integration) aligns closely with GWT’s macroscale architecture. The distinctive addition here is the claim that microtubule-level dynamics may modulate *which* signals gain access to the global workspace, effectively adding a subcellular filtering layer below GWT’s neural-network level of description.

Key distinction: The present hypothesis is framework-agnostic at the macroscale: it addresses a level of organisation that neither IIT nor GWT currently models. Unlike both theories, which remain agnostic about subcellular mechanisms, this hypothesis makes specific empirical commitments about microtubule biophysics that are testable and falsifiable (Section 7). Conversely, unlike Orch OR (Section 1.3), it does not invoke quantum gravity or claim that microtubules are the *seat* of consciousness—only that they may modulate neural dynamics relevant to conscious processing.

8.4 Tier Summary

9 Limitations and Open Questions

- (1) The UV superradiance evidence (Tier 1) and the 39 Hz electrical oscillation evidence (Tier 2) are currently independent observations in the same structure. The critical bridge experiment—demonstrating that optical perturbation of tryptophan superradiance causally modulates the 39 Hz ionic oscillation—has not been performed.
- (2) All microtubule electrophysiology data are from *in vitro* preparations. Whether identical oscillatory dynamics occur in intact neurons in living brain tissue, and whether they causally contribute to EEG gamma beyond classical synaptic models, remains undemonstrated.
- (3) Lithium isotope evidence is promising but contested. Some mechanistic probes show no isotope selectivity at membrane transport sites. Broad independent replication is lacking.
- (4) The epothilone B studies, while statistically robust, involve small samples ($n = 12$ rats in the original study, plus a mouse replication and a *Tetrahymena* study). A parsimonious alternative remains binding-site competition. A chronic dosing study found drug-specific directional effects, complicating interpretation [23].
- (5) The philosophical extension (universal consciousness field) is not testable with current methods and must not be conflated with the empirical content.

- (6) Classical microtubule functions (axonal transport, synaptic scaffolding, morphological maintenance) provide alternative explanations for some observed effects of microtubule perturbation. Any experiment claiming to demonstrate quantum-specific consciousness effects must control for these classical roles.
- (7) The Craddock et al. (2017) computational prediction that anaesthetics specifically shift a 613 THz collective dipole mode in tubulin has not been independently replicated or experimentally validated by terahertz spectroscopy. Its robustness remains unknown.
- (8) No study has directly measured the effect of microtubule-targeting drugs on EEG or LFP gamma-band power in slice or *in vivo* preparations designed to isolate microtubule contributions from classical synaptic mechanisms. This represents the single most important experimental gap in the framework.
- (9) The quantitative plausibility simulations (Section 6) are computational exercises exploring parameter space, not fits to *in vivo* data. Simulation 2 uses a time-scale calibration that makes the reference frequency (39.1 Hz) circular by construction; only the parameter sweep results are genuinely predictive. The inter-soliton coupling $\kappa = 0.63$ and the Wilson–Cowan attenuation factor (5×10^{-4}) are constrained free parameters without independent experimental sources. The Wilson–Cowan analysis (Section 6.3) shows that the isolated microtubule aggregate signal ($\sim 63 \mu\text{V}$) is insufficient to modulate network gamma without additional amplification mechanisms whose *in vivo* existence remains unconfirmed. This bottleneck should be regarded as the principal quantitative weakness of the hypothesis.
- (10) This framework focuses exclusively on neuronal microtubules. Astrocytes and oligodendrocytes also contain extensive microtubule networks and are increasingly recognised as active modulators of synaptic transmission and network synchrony. Whether glial microtubules exhibit similar oscillatory dynamics, and whether they contribute to or confound the proposed neuronal mechanism, remains entirely unexplored.

10 Conclusion

This framework should be understood as a **multiscale consciousness hypothesis** with asymmetric evidential support across levels. The thalamocortical component is strongly grounded in contemporary clinical and systems neuroscience. The microtubule component is biologically interesting and experimentally nontrivial, but still indirect with respect to conscious experience.

Three quantitative plausibility simulations establish that: (i) the aggregate microtubule voltage under

Table 1: Summary of the three-tier evidential framework.

Tier	Domain	Key Evidence	Status	Critical Gap	Proposed Test
3	Thalamocortical	PCI, DBS, lesion data	<i>In vivo</i> , clinical	No single decisive gap; prefrontal role debated	Expts 2, 3
2	MT electrodynamics	39 Hz oscillation, soliton model	<i>In vitro</i>	<i>In vivo</i> validation	Expts 1, 7
1	MT quantum optics	UV superradiance, energy migration	<i>In vitro</i>	Tier 1→2 bridge	Expt 6

incoherent summation ($\sim 63 \mu\text{V}$) is small but scales predictably with coherence and bundle amplification; (ii) the 39 Hz oscillation frequency is robust under parameter perturbation in the coupled nonlinear transmission line model, given the calibrated time-scale anchor; and (iii) resonant entrainment of cortical gamma oscillations is possible under the bundle-amplified scenario but not under the conservative isolated estimate, identifying the amplification question as the critical empirical bottleneck.

The most important unresolved issue is the absence of a demonstrated causal bridge from microtubule quantum-optical or electrodynamic states to *in vivo* neural integration measures such as gamma synchrony, PCI, or conscious report. Until that bridge is experimentally established, the model remains a productive but unconfirmed hypothesis.

The seven proposed experiments provide a concrete empirical programme. If the predictions are confirmed, microtubules would be established as a functionally relevant layer in the neural architecture of consciousness, operating below and in concert with thalamocortical integration. If the predictions fail, the hypothesis would be substantially weakened and the field’s attention rightly directed elsewhere.

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